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Genetic silencing of AKT induces melanoma cell death via mTOR suppression

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Abstract

Aberrant activation of the PI3K-AKT pathway is common in many cancers including melanoma, and AKT1, 2 and 3 (AKT1–3) are *bona fide* oncoprotein kinases with well-validated downstream effectors. However, efforts to pharmacologically inhibit AKT have proven to be largely ineffective. In this study, we observed paradoxical effects following either pharmacologic or genetic inhibition of AKT1–3 in melanoma cells. Although pharmacological inhibition was without effect, genetic silencing of all three AKT paralogs significantly induced melanoma cell death through effects on mTOR. This phenotype was rescued by exogenous AKT1 expression in a kinase dependent manner. Pharmacological inhibition of PI3K and mTOR with a novel dual inhibitor effectively suppressed melanoma cell proliferation *in vitro* and inhibited tumor growth *in vivo*. Furthermore, this single agent targeted therapy was well tolerated *in vivo* and was effective against MAPK inhibitor resistant patient-derived melanoma xenografts. These results suggest that inhibition of PI3K and mTOR with this novel dual inhibitor may represent a promising therapeutic strategy in this disease in both the first line and MAPK inhibitor resistant setting.

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AUTHOR CONTRIBUTIONS

GP, MM, and SH designed experiments. GP, TT, DK, WB, CS, KO, RF, AC, RE, KS, and MF performed experiments. GP, DK, WB, CS, MM, and SH analyzed the data. GP, MM and SH wrote the manuscript. All authors discussed the results and reviewed and revised the manuscript accordingly.

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Keywords

Melanoma; AKT; PI3K; mTOR; targeted therapy

INTRODUCTION:

Approximately fifty-percent of all melanomas harbor a *BRAF*^{T1799A} mutation that encodes the BRAF^{V600E} oncoprotein kinase, the expression of which leads to constitutive activation of MAP kinase signaling. The importance of this pathway in melanoma maintenance has been emphasized by the FDA approval of drugs that target BRAF^{V600E} and its downstream effector MEK for the treatment of *BRAF*-mutated melanoma (1). However, in both preclinical models and human melanocytes, expression of BRAF^{V600E} drives the development of benign melanocytic neoplasia that generally fails to progress to melanoma without additional cooperating alterations (2). While the mechanism(s) by which BRAF^{V600E} cooperates with other alterations to convert normal melanocytes to metastatic melanoma cells has yet to be fully elucidated (3,4), aberrant activation of phosphatidylinositol 3'-lipid (PI3'-lipid) signaling drives the progression of BRAF^{V600E}-driven melanoma in genetically engineered mouse (GEM) models (5). Moreover, constitutive activation of PI3'-lipid signaling is observed widely across human malignancies, including melanoma, either due to silencing of the tumor suppressor PTEN, which is a PI3'-lipid phosphatase, or mutational activation of *PIK3C* genes encoding PI3'-kinases (e.g., *PIK3CA* encoding PI3'-kinase-α). Of the various known effectors of PI3'-lipid signaling, the AKT (AKT1, 2, and 3) family of protein kinases are thought to play a prominent role in melanomagenesis (6). In normal cells, AKT protein kinases are activated downstream of numerous growth factor receptors that activate PI3'-lipid signaling. However, in cancer cells, activating mutations or copy number gains of *AKT* genes have been reported in samples from melanoma patients (7). The three highly homologous AKT paralogs typically reside in the cytoplasm in an inactive conformation. Though, in response to the increased activity of PI3'-kinases, they bind via their N-terminal pleckstrin homology (PH) domains to PI3'-lipids at the cell membrane, where they undergo activating phosphorylation by PDK1 and mTORC2 (8). Once activated, AKT protein kinases phosphorylate a multitude of target proteins that in turn regulate cell survival, growth, motility, and invasion (9). In melanoma, the level of phospho-AKT has been reported to steadily increase throughout the progression from dysplastic nevi to metastases. Moreover, silencing of PTEN or mutational activation of AKT has been reported as a mechanism of acquired resistance to FDA-approved inhibitors of BRAF^{V600E} signaling in *BRAF*-mutated melanoma, highlighting the potential importance of targeting PI3K/AKT signaling for the treatment of this disease (10).

In preclinical models, the inhibition of PI3K signaling has a cytostatic effect on melanoma cell proliferation. In contrast, pharmacological inhibition of AKT, including the use of structurally unrelated and mechanistically dissimilar agents, has little to no effect on melanoma cell proliferation (11). Furthermore, clinical trials evaluating AKT inhibitors as single agents have shown minimal efficacy in melanoma, suggesting that AKT may not be essential for melanoma cell survival (12). However, these findings are complicated by the relief of negative feedback signaling mediated by mTORC1 and/or p70^{S6K}, which

leads to the activation of the PI3K signaling pathway following AKT inhibition (13). Our previous research indicated that the effects of PI3K inhibitors on the activity of mTORC1, leading to the regulation of protein synthesis, are independent of AKT (14). Moreover, additional PI3K-dependent protein kinases, such as the serum and glucocorticoid-regulated kinase (SGK) 1–3 family, have been identified and are known to have downstream effectors similar to AKT (15). Finally, although PI3'-kinase inhibition forestalled the onset of MEK inhibitor resistance in BRAF^{V600E}-driven melanoma (11), translation of such observations to routine clinical use has been hampered by the consequent toxicity of combined inhibition of BRAF^{V600E} and PI3'-kinase signaling (16,17). Hence, these data highlight the critical need to understand the importance of PI3K/AKT signaling in melanoma maintenance as a step towards more effective combination pathway-targeted therapy in the subset of melanomas in which these pathways cooperate for melanoma proliferation and survival.

In this study, we evaluated the role of AKT in melanoma cell proliferation and observed a striking difference in melanoma cell viability after pharmacological inhibition compared with RNA interference (RNAi)-mediated silencing of total AKT expression. We noted that a combination of three small interfering RNAs (siRNAs) targeting AKT1, 2, and 3 (henceforth referred to as siAKT1–3) resulted in potent melanoma cell lethality, which was rescued by the expression of siRNA-resistant forms of AKT. Interestingly, genetic suppression of AKT1–3 significantly decreased mTORC activity, whereas pharmacological inhibition of AKT did not. However, use of newer-generation mTOR inhibitors, notably a dual PI3K/mTOR inhibitor, led to a more effective decrease in melanoma cell proliferation as well as inhibition of melanoma growth *in vivo* and increased overall survival of treated mice. These data emphasize the complexity of PI3'-lipid signaling in melanoma and indicate that while pharmacological blockade of AKT is unlikely to be a successful strategy for the treatment of *BRAF*-mutated melanoma, dual targeting of PI3K and mTOR may be an effective alternative approach.

MATERIALS AND METHODS

Viral constructs and propagation

The avian retroviral vectors used in this study are replication-competent Avian Leukosis Virus splice acceptor and Bryan polymerase-containing vectors of envelope subgroup A [designated RCASBP(A) and abbreviated RCAS]. A gateway-compatible RCAS destination vector has been described previously (18). RCAS-Cre and RCAS-myrAKT1 have also been previously described (19). RCAS-myrAKT1 was used as a template to generate the *Akt1* phosphorylated mutant constructs using Q5[®] Site-Directed Mutagenesis (New England Biolabs). Myr-Akt was mutated using the following primers:

S473A FWD “CACTTCCCCCAGTTCGCCTACTCAGCCAGTGGC,”
 S473A REV “GCCACTGGCTGAGTAGGCGAACTGGGGGAAGTG,”
 T308A FWD “TGCCACTATGAAGGCATTCTGCGGAACGCCG,” T308A
 REV “CGGCGTTCCGCAGAATGCCTTCATAGTGGCA.” S473D FWD
 “CACTTCCCCCAGTTCGACTACTCAGCCAGTGGC,” S473D REV
 “GCCACTGGCTGAGTAGTTCGAACTGGGGGAAGTG,” T308D FWD

“TGCCACTATGAAGGATTTCTGCGGAACGCCG,” and T308D REV
“CGGCGTTCGCGAGAAATCCTTCATAGTGGCA.”

pDONR223-SGK (wild-type) was obtained from Addgene and recombined into the destination vectors pDEST FG12_CMV_GW and SV40_Luciferase-IRES-eGFP. pDONR223-SGK was used as a template to generate the myr SGK1 construct and SGK1 kinase-dead (K127M) constructs using Site-Directed Mutagenesis (New England Biolabs). SGK1 was mutated using the following primers: K127M FWD, CTATGCAGTCATGGTTTTACAGAAGAAAG; K127M REV, AACACTTCTTCTGCCTTG.

Cell culture

The melanoma cell lines A375, HT144, SK-MEL28, WM793, YUMM3.2 (Yale University Mouse Melanoma 3.2) were cultured in DMEM/F12 media (Thermo Fisher) supplemented with 10% FBS (Atlanta Biologicals, Flowery Branch, GA) and maintained at 37°C in 5% CO₂. DF-1 cells were grown in DMEM-high glucose medium (Thermo Fisher) supplemented with 10% FBS (Atlas Biologicals) and 0.5 µg/mL Gentamicin (Thermo Fisher), and maintained at 39°C. The patient-derived cell lines MTG001 and MTG004 were grown in MEL-2 medium consisting of 80% MCDB153 (Sigma), 20% L15 (Gibco), 2% FBS (Denville), 1.68 mM CaCl (Sigma), 1× insulin (Gibco) 5 ng/ml EGF (Sigma), 15 µg/ml Bovine Pituitary Extract (Gibco), and 1× Pen/strep (Gibco) and maintained at 37°C in 5% CO₂. All established human cell lines used for these studies have been authenticated by STR profiling and mycoplasma testing by PCR is completed quarterly.

Viral infections (*in vitro*)

A375, HT144, and WM793 cells were transfected with pcDNA3.1-TVΔ (20) containing the hygromycin B resistance gene to generate A375-TVA, HT144-TVA, and WM793-TVA cells. TVA-positive clones were selected using 300 µg/mL Hygromycin B (Thermo Fisher). Supernatants from DF-1 cells producing RCAS myr-HA-Akt1-T308A, RCAS myr-HA-Akt1-T308D, RCAS myr-HA-Akt1-S473A, myr-HA-Akt1-S473D, myr-HA-Akt1-T308A, S473A, myr-HA-Akt1-T308A, S473D, myr-HA-Akt1-T308D, S473A, myr-HA-Akt1-T308D, S473D, myr-HA-Akt1-delta(PH), and myr-HA-Akt1-delta(PH)-K179M were used to infect A375, HT144, and WM793 TVA+ cells. The expression of Akt1 mutants in A375-TVA+, HT144-TVA+, and WM793-TVA+ cells was confirmed by immunoblotting for HA. To generate the YUMM 3.2 SGK1 isogenic cell line, myr-SGK1 was gateway cloned from pDONR221 TOPO to an FG12 CMV-luciferase-IRES-eGFP destination vector to generate FG12 CMV myr-SGK1; luciferase-IRES-eGFP. The lentiviral vector FG12 CMV myr-SGK1;Luciferase-IRES-eGFP, along with packaging plasmids psPAX2 (#12260 Addgene, Cambridge, MA, USA) and pCMV-VSV-G (#8454 Addgene) were transfected into 293FT cells using the calcium phosphate method. Supernatants from these virus-containing cells were then used to infect isogenic YUMM 3.2 parental and Pten^{-/-} YUMM3.2 SGK1 cells, which were sorted for eGFP using a Propel Labs Avalon at the Flow Cytometry Shared Resource Laboratory at the University of Utah.

Immunoblotting

Protein extracts were harvested by lysing the cells in radioimmunoprecipitation assay (RIPA) lysis buffer containing Halt Protease and Phosphatase Inhibitor Cocktail (Thermo Scientific). A BCA assay was performed to determine the protein concentration prior to boiling. Samples were standardized to equal concentrations and diluted in NuPAGE LDS (lithium dodecyl sulfate, pH 8.4) sample loading buffer (Invitrogen), prior to boiling for 10 min at 95°C. Proteins were separated using standard SDS-PAGE gel electrophoresis with 8–16% gradient Tris-Glycine polyacrylamide gels, transferred to nitrocellulose membranes for immunoblot analysis using an iBlot2 (Invitrogen), and stained with primary antibodies, as indicated in as indicated in each figure. LI-COR brand Intercept (TBS) blocking buffer, IRDye RD and 800CW secondary antibodies were used for all immunoblotting experiments except Figure 6C, which used 5% BSA/TBS-Tween (TBS-T) or 5% non-fat dry milk (NFDM)/TBS-T blocking buffer and HRP conjugated anti-mouse or anti-rabbit secondary antibodies (Cell Signaling Technology) as appropriate. The membranes were scanned and analyzed using the Odyssey Imaging System and software (LI-COR) for all experiments except Figure 6C, which was scanned using the Azure Imaging system (Azure Biosystems). Antibodies used: HA (MMS-101P; Biolegend), GAPDH (sc-47724; Santa Cruz Biotechnology), β -actin (3700; Cell Signaling Technology), AKT (4691; Cell Signaling Technology), Phospho-AKT (T308) (13038; Cell Signaling Technology), Phospho-AKT (S473) (3787; Cell Signaling Technology), PRAS40 (2691; Cell Signaling Technology), Phospho-PRAS40 (Y246) (2997; Cell Signaling Technology), PTEN (9188; Cell Signaling Technology), Phospho-p44/42 MAPK (4370; Cell Signaling Technology), p44/42 MAPK (9102; Cell Signaling Technology), Phospho-S6 ribosomal protein (Ser235/235) (4858; Cell Signaling Technology), S6 ribosomal protein (2217; Cell Signaling Technology), Akt1 (2938; Cell Signaling Technology), Akt2 (3063; Cell Signaling Technology), Akt3 (3788; Cell Signaling Technology), Phospho-4E-BP1 (T37/46) (2855; Cell Signaling Technology), 4E-BP1 (9644; Cell Signaling Technology); SGK1 (D72C11; Cell Signaling Technology); Phospho-NDRG1 (Thr348) (D98G11; Cell Signaling Technology); Caspase 3 (3G2) (9668; Cell Signaling Technology); Cleaved Caspase-3 (D175) (9661; Cell Signaling Technology).

Pharmacological inhibitors

AKT inhibitors (MK2206 and GSK2141795), PI3K inhibitors (GDC-0941 and BYL719), mTORC inhibitor (RapaLink-1), and SGK inhibitor (GSK650394) were purchased from Selleck Chemicals (Houston, TX, USA). AKT inhibitor GDC-0068 was generously provided by Genentech. Paxalisib was generously provided by Kazia Therapeutics (21). AKT PROTAC (INY-03–041) was kindly provided by Dr. Nathanael S. Gray (Stanford University) and Dr. Alex Toker (Harvard Medical School). Each drug was formulated in DMSO and added to F12/DMEM (Thermo Fisher) at a final concentration of 0.1% (v/v) DMSO. Inhibitors were used at selected concentrations, as indicated in the figures and text. Chemical structures of all pharmacological agents used in this study are provided in Supplementary Figure S7.

Incucyte assays

Cell proliferation was assessed by seeding ~3,000 cells (A375 and WM793), ~5,000 cells (HT144 and SK-MEL28), or ~7,000 cells (MTG001 and MTG004) per well in 96-well plates. Experiments were performed on two to three separate occasions in triplicate wells for each experimental condition. Pharmacological agents were added 24 hours after plating. Cells were cultured in the absence or presence of pharmacological agents for 3–5 days, or until the control reached ~100% confluency. Cell death was assessed using the Incucyte® Cytotox Red and Incucyte® Caspase 3/7 assays per the manufacturer's specifications. Confluence or cell death was assessed over time using an IncuCyte® Zoom Live Cell Imaging instrument with data analyzed using IncuCyte® Analysis Software (Sartorius) at two-hour intervals.

Luminex assay

Fluorescence bead-based measurements were carried out on a Luminex MAGPIX® according to the manufacturer's instructions using the following kit: Bio-Plex Pro Cell Signaling Akt panel, 8-plex (LQ00006JK0K0RR).

Lipofectamine-mediated siRNA transfection

siRNAs against human AKT1 (J-003000–10-0005 and J-003000–11-0005), AKT2 (J-003001–09-0005 and J003001–10-0005), AKT3 (J-003002–13-0005 and J-003002–14-0005), SGK1 (J-003027–13-0005), SGK2 (J-004673–09-0005), SGK3 (J-004162–06-0005), and a negative control siRNA with a scrambled sequence were purchased from Dharmacon (and transfected with Lipofectamine RNAiMAX reagent at a total concentration of 50nM per siRNA-treated condition). To test for rescue of the effects of siAKT1–3, constructs encoding mouse AKT protein kinases, which are resistant to the human-specific siRNAs described above, were generated and expressed in human melanoma cells, and their expression was confirmed by immunoblotting. Rescue assays were performed in media supplemented with 10%(v/v) FBS, as previously described, using Cell Titer 96 MTS (Promega). Briefly, cells were seeded at ~50% confluence on the day prior to transfection. Lipofectamine-mediated siAKT1–3 transfection (as described above) was performed at time point 0, and an MTS assay was performed 48 h post-transfection. Each assay was performed in triplicate, with at least three independent technical replicates. Data were plotted and analyzed using the GraphPad Prism Software.

Phospho-PRAS40 enzyme-linked immunosorbent assays (ELISA)

The levels of phosphorylated proline-rich AKT substrate of 40kDa (phospho-PRAS40 (pT246)) were assessed using PRAS40 ELISA kits (Invitrogen, Camarillo, USA, KHO0421). Cells were lysed in extraction buffer (Invitrogen, Camarillo, CA, USA, FNN0011), and samples were diluted 1:50 in diluent buffer. Absorbance was measured at 450 nm upon completion of the assay.

In vivo tumorigenesis studies

Four-to eight-week-old immune-competent mice tolerized to EGFP and luciferase were injected subcutaneously into the right flank with 2.5×10^5 melanoma cells and observed for

tumorigenic growth. Tumors were visualized and measured weekly using bioluminescence imaging, and tumor burden was quantified using luminoscope values and digital caliper measurements. The following formula was used to calculate tumor volume: $(\text{Length} \times \text{Width}^2)/2$. All mouse tissues were fixed in 10% neutral buffered formalin overnight and dehydrated in 70% ethyl alcohol.

***In vivo* patient derived xenografts (PDX)**

The University of Utah IRB (protocols 89989 and 10924) approved human sample collection following informed consent. The clinical information in this study is deidentified and published in accordance with the ethics approvals for this study. Melanoma Patient Derived Xenograft (PDX) models were created by implanting viable human tumor fragments (~2–3 mm) subcutaneously through a small incision on the dorsal region of four-to-six week old immune-compromised NOD *scid* gamma (NSG; NOD.Cg-Prkdc^{scid} Il2rg^{tm1Wjl}/SzJ), Jackson Laboratory stock 5557, or NOD *rag* gamma (NRG, NOD-*Rag1*^{null} *IL2rg*^{null}, NOD *rag* gamma, NOD-RG) mice, Jackson Laboratory stock 7799. Mice were monitored for health and tumors were measured weekly with digital calipers once growth was observed.

RT-PCR

Total RNA from MTG001 and MTG004 PDX-derived cell lines was isolated using Qiagen RNA Isolation and Purification. RNA was converted to cDNA using iScript Reverse Transcription Supermix (BioRad). cDNA was analyzed by SYBR green real-time PCR with SsoAdvanced Universal SYBR Green Supermix (BioRad). Gene expression was calculated relative to *Actin* using the $2^{-(\text{dCt.x} - \text{average}(\text{dCt.control}))}$ method and normalized to the control group for graphing. Primer sequences used are listed as follows: *Actin* FWD “CGTCTTCCCCTCCATCGT”, *Actin* REV “GAAGGTGTGGTGCCAGATTT”, *AKT1* FWD “ATGAGCGACGTGGCTATTGTGAAG”, *AKT1* REV “GAGGCCGTCAGCCACAGTCTGGATG”, *AKT2* FWD “ATGAATGAGGTGTCTGTTCATCAAAGAAGGC”, *AKT2* REV “TGCTTGAGGCTGTTGGCGACC”, *AKT3* FWD “ATGAGCGATGTTACCATTGT”, and *AKT3* REV “CAGTCTGTCTGCTACAGCCTGGATA.”

RNA sequencing and mutation detection

For RNA extraction and DNase treatment, the Roche High Pure miRNA Isolation Kit was used in accordance with the manufacturer’s instructions (5080576001). RNA sequencing was performed at the High-Throughput Genomics and Bioinformatics Analysis Core at the University of Utah. RNA libraries were prepared using the Illumina TruSeq Stranded Total RNA Kit with Ribo-Zero Gold. Total RNA samples (100–500 ng) were hybridized with Ribo-Zero Gold to substantially deplete cytoplasmic and mitochondrial rRNA from the samples. Stranded RNA sequencing libraries were prepared using the Illumina TruSeq Stranded Total RNA Kit with Ribo-Zero Gold per the manufacturer’s instructions (RS-122–2301 and RS-122–2302). Purified libraries were qualified on an Agilent Technologies 2200 TapeStation using a D1000 ScreenTape assay (cat# 5067–5582 and 5067–5583). The molarity of adapter-modified molecules was defined by quantitative PCR using the Kapa Biosystems Kapa Library Quant Kit (cat#KK4824). Individual libraries were normalized to 10 nM and equal volumes were pooled in preparation for Illumina sequence analysis.

To assess mutation status of specific transcripts in patient-derived tumor samples, BAM files were formatted with read groups assigned using AddOrReplaceReadGroups from Picard; then mutations were called using Mutect2 from the GATK suite, followed by FilterMutectCalls from GATK. Finally, variants with an appreciable allele frequency (>0.3 with a depth of at least five reads) were scanned against the ClinVar database to ascertain if they were classified as benign, pathogenic, or of uncertain significance.

***In vivo* drug treatments**

For *in vivo* drug testing, drug treatment (or vehicle control treatment) was initiated when tumors reached ~ 100 – 200 mm^3 in size. Mice were randomized into treatment groups based on the establishment of equal tumor size and sex per group. Tumor size was measured twice weekly using digital calipers, and tumor volume was calculated using the following formula ($\text{length} \times \text{width}^2/2$). All dosing regimens were performed five days a week at the dose and time period indicated. Drugs were dissolved in 0.5% carboxymethylcellulose + 0.2% Tween-80 dissolved in water as the vehicle and delivered by oral gavage.

Bioluminescence imaging

Mice were injected intraperitoneally with 16.7 mg/mL D-Luciferin in 200 μL of phosphate buffered saline 10 min prior to image acquisition. The IVIS Spectrum was used to acquire images at one-week intervals beginning one week after subcutaneous implantation of mouse YUMM3.2 melanoma cells until the experimental endpoint (6 weeks post-injection) or when the tumor burden reached 2000 mm^3 . Living Image software (version 4.5.2) was used to compile images.

Statistical Methods

For most experiments, P-values were determined by two-tailed T-tests. Densitometry measurements were performed using ImageJ, and protein levels were normalized as phosphorylated to total protein and/or GAPDH. The data are represented as mean $\pm$ S.E.M. Mouse censored survival data were analyzed using the log-rank (Mantel-Cox) test of the Kaplan-Meier estimate of survival.

Study Approval

All animal experiments were performed at AAALAC-approved facilities at the University of Utah. All animal protocols were reviewed and approved prior to experimentation by the Institutional Animal Care and Use Committee (IACUC) at the University of Utah.

Ethical Compliance

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Data Availability

The data generated in this study are available upon request from the corresponding author.

RESULTS

siRNA-mediated knockdown of all three AKT paralogs (siAKT1–3) is lethal to melanoma cells

We first tested the possible importance of AKT in melanoma cell proliferation *in vitro* using multiple complementary approaches. Our previous data indicated that pharmacological AKT inhibitors had little to no effect on melanoma cell proliferation, despite their confirmed effects on phosphorylated AKT (14). To validate this further, we tested the effect of pharmacological blockade of AKT with two mechanistically dissimilar compounds: the pan-AKT allosteric inhibitor MK2206 (AKTi¹) and pan-AKT ATP competitive inhibitor GDC-0068 (AKTi²) on the proliferation of multiple *BRAF*-mutated melanoma cell lines (A375, HT144, SK-MEL28, and WM793) (22,23). Consistent with previous results, although A375, HT144, WM793, and SK-MEL28 cells were sensitive to inhibition of all four class 1 PI3Ks with GDC0941, we observed no significant effect of AKT inhibition on melanoma cell proliferation regardless of PTEN status (Supplementary Figure S1). Thus, pharmacological inhibition of AKT using two different compounds in a variety of cell lines had no observable effect on melanoma cell proliferation *in vitro*. In addition to the four commercially available cell lines listed above, we also evaluated two cell lines derived from *BRAF*^{V600E}-driven melanoma patient-derived xenografts (MPDX), referred to here as MTG001 and MTG004. These cell lines displayed detectable levels of pAKT, PTEN, pERK and p-Ribosomal S6 Kinase as assessed by immunoblotting (Supplementary Figure S2). Pharmacological inhibition of AKT using two different compounds in patient-derived cell lines also had no observable effect on melanoma cell proliferation *in vitro* (Supplementary Figure S1).

Because pharmacological inhibition was largely unaffected, we attempted to silence AKT expression in *BRAF*-mutated melanoma cells using RNA interference. To this end, we obtained two siRNAs targeting each of the three human AKT paralogs. A375, HT144, SK-MEL28, WM793, MTG001, and MTG004 cells were transfected with each set of siRNAs as well as combinations thereof (siAKT1, siAKT2, siAKT3, siAKT1+2, siAKT1+3, siAKT2+3, and siAKT1+2+3), and cell confluence was assessed as a measure of proliferation (Figure 1A). To validate the specificity of the siRNAs, we confirmed the knockdown using antibodies specific for AKT1, AKT2, and AKT3 or RT-PCR with paralog specific primers (Figure 1B and Supplementary Figure S3). Silencing of individual AKT paralogs had no effect on cell proliferation when compared to a non-targeting siRNA control. Pairwise knockdown of AKT paralogs 1 and 3 resulted in a statistically significant decrease in cell proliferation in three of the cell lines tested, HT144, SK-MEL28, and MTG004 ($p < 0.0001$), but the other pairwise combinations had no effect. However, most strikingly, siAKT1–3 resulted in complete inhibition of proliferation at approximately 24 h in all cell lines tested (Figure 1A; $p < 0.0001$). This was accompanied by significantly decreased levels of phosphorylated AKT at serine 473 (pS473) 24 or 26 h post-transfection and significantly decreased total AKT protein expression at 26 h (Figure 1C). To confirm the on-target specificity of the observed siRNA effects, we performed a rescue experiment using mouse AKT expression vectors that were resistant to human-specific AKT1–3 siRNAs. We first validated the specificity of siAKT1–3 for human AKT in YUMM1.1 mouse

melanoma cells. As expected, the siRNAs were human specific and had no effect on AKT expression in YUMM1.1s (Supplementary Figure S4). We then transiently co-transfected *BRAF*-mutated human cells with siAKT1–3 and vectors encoding mouse AKT1, AKT2, or AKT3. We found that the expression of mouse AKT1 could rescue the effects of the siAKT1–3 phenotype in human melanoma cells (Figure 1C–D). These data indicate that the anti-proliferative effects of siAKT1–3 are on-target and that any one paralog of mouse AKT can compensate for the other two.

To determine whether the observed decrease in cell proliferation caused by siAKT1–3 was due to the induction of cell death or inhibition of cell growth, we performed a cell death assay. While a pleiotropic kinase inhibitor control, staurosporine, induced a gradual increase in the percentage of dead cells over several hours, siAKT1–3 caused a dramatic increase in cell death at approximately 22 h, which was abrogated by ectopic expression of mouse AKT1 (Figure 1D). Together, these data demonstrated that melanoma cell death is induced only when all three human paralogs of AKT are genetically silenced.

Next, we assessed the mechanism of cell death upon siAKT1–3 knockdown. To evaluate this, we employed caspase inhibitors with varying specificities, including Z-VAD-FMK, a pan-caspase inhibitor, Z-DEVD-FMK, caspase 3 inhibitor, and Z-IETD-FMK, caspase 8 inhibitor (24). A375 and MTG004 cells were transfected with siAKT1–3 and treated with 50 μ M of each caspase inhibitor two hours later. Compared to siAKT1–3 alone, co-treatment with any of the caspase inhibitors protected cells from death, indicating both intrinsic and extrinsic apoptosis as the mechanism of siAKT1–3-induced cell death (Supplementary Figure S5A). Furthermore, siAKT1–3 led to an increase in MTG004 cell death, a phenotype that was rescued by co-treatment with each caspase inhibitor (Supplementary Figure S5B–C).

Recently, an AKT proteolysis-targeting chimera (PROTAC), INY-03–041 was described. This pan-AKT degrader consists of GDC-0068 conjugated to lenalidomide, a recruiter of the E3 ubiquitin ligase cereblon (25). INY-03–041 has a dual mechanism of action that inhibits AKT protein kinase activity (through the GDC-0068 moiety) and targets AKT protein for proteasomal degradation (through CRBN-mediated ubiquitination). Therefore, we hypothesized that it would induce melanoma cell death similar to siAKT1–3. In MTG001 and MTG004 cells, 1 μ M INY-03–041 significantly reduced melanoma cell proliferation but not as potently as siAKT1–3 (Supplementary Figure S6A). We observed a ~77% reduction in total AKT protein expression following INY-03–041 treatment, compared with a ~89% reduction by siAKT1–3 (Supplementary Figure S6B). Whereas siRNAs targeting AKT1–3 led to almost complete loss of each of the total paralogs (Figure 1B), time-course immunoblotting of cell lysates treated with varying concentrations of INY-03–041 demonstrated incomplete total protein knockdown of individual AKT paralogs (Supplementary Figure S6C–D). Although not as effective as siAKT1–3, the efficacy of INY-03–041 further supports the hypothesis that AKT is required for melanoma cell proliferation and survival.

Rescue of melanoma cells from siAKT1–3 is dependent on AKT kinase activity

AKT has been reported to have both kinase-dependent and kinase-independent functions. Some of the kinase-independent functions are mediated by the PH domain, which binds to PI3'-lipids and influences downstream signaling (26). Having shown that mouse AKT rescued the effects of siAKT1–3, we tested whether this rescue was dependent on AKT protein kinase activity. To this end, we generated a kinase-inactive form of mouse AKT1 in which lysine 179 was substituted with methionine to render the enzyme catalytically inactive. This mutant also harbors a deletion of the PH domain (amino acids 11–60, ΔPH) to eliminate any possible dominant-negative effects mediated by the sequestration of PI3'-lipids. A myristoylation tag (amino acids 1–14 of c-SRC, myr) was added to ensure that myr-HA-ΔPH-AKT1^{K179M} was targeted to the plasma membrane for activation in the absence of a functional PH domain and an HA tag was added for immunoblot detection. A375, HT144, and WM793 cells engineered to express myr-HA-ΔPH-AKT1^{K179M}, were treated with siAKT1–3, and cell viability was assessed 48 h after siRNA transfection. Expression of myr-HA-ΔPH-AKT1^{K179M} failed to rescue cell viability in the presence of siAKT1–3, whereas a catalytically active form of this protein (myr-HA-ΔPH-AKT1) demonstrated a robust rescue in this context (Figure 2A). This suggests that functional kinase activity rather than kinase-independent signaling is required to maintain melanoma cell viability.

To further understand the importance of AKT phosphorylation in melanoma cell survival, we investigated the role of two major phosphorylation sites, T308 and S473, in the rescue of siAKT1–3-mediated cell death. Numerous phosphorylation sites have been identified on AKT of which three: T308, T450 and S473 have thus far been regarded as being most important for full AKT kinase activity. T308 is phosphorylated by PDK1 and S473 by mTORC2, and T450 is thought to be a site of AKT autophosphorylation (27,28). Using site-directed mutagenesis, we generated S/T>A (phospho-deficient) and S/T>D (phospho-mimetic) alterations at T308 and S473 in myr-HA-AKT1. A375, HT144, and WM793 cells stably expressing these various forms of AKT1 were generated, and the cell viability of each cell line treated with siAKT1–3 was measured 48 h post-transfection, as described above. As expected, the two constitutively active forms of AKT, myr-HA-AKT1^{S473D} and myr-HA-AKT1^{T308D}, rescued the cell viability. Moreover, phospho-deficient myr-HA-AKT1^{S473A} also rescued the cell viability. In contrast, phospho-deficient myr-HA-AKT1^{T308A} failed to rescue this effect (Figure 2A). These data emphasize the critical importance of T308 phosphorylation for AKT1 kinase activity.

To evaluate the ability of individual phospho-mimetic and phospho-deficient mutants to cooperate with each other, we generated double point mutants: 1. myr-HA-AKT1^{T308A/S473A}, 2. myr-HA-AKT1^{T308A/S473D}, 3. myr-HA-AKT1^{T308D/S473A}, and 4. myr-HA-AKT1^{T308D/S473D}. As expected, the ability of these constructs to rescue the effects of siAKT1–3 was dependent on T308, as any construct with T308A failed to fully rescue (Figure 2B). Interestingly, myr-HA-AKT1^{T308D/S473A} only achieved ~50% rescue, despite the ability of individual mutants to fully rescue the phenotype. To evaluate the ability of these various forms of AKT to promote phosphorylation of a direct AKT substrate, we used an ELISA assay to assess pT246-PRAS40 in cell extracts. As expected, pT246-

PRAS40 was largely undetectable in cells expressing myr-HA- Δ PH-AKT1^{K179M}, myr-HA-AKT1^{T308A/S473D} or myr-HA-AKT1^{T308A/S473A} in accordance with the inability of these mutants to rescue siAKT1–3 (Figure 2C). In contrast, pT246-PRAS40 was detected in cells in which various AKT1 constructs successfully rescued the effects of siAKT1–3: myr-HA- Δ PH-AKT1, myr-HA-AKT1^{T308D/S473A}, and myr-HA-AKT1^{T308D/S473D}.

Genetic silencing of AKT significantly decreases mTORC activity

To identify downstream signaling effectors that lead to cell death upon treatment with siAKT1–3, we utilized a fluorescence-based multiplex assay to measure the phosphorylation of a panel of cell signaling molecules in the PI3K>AKT>mTORC signaling pathway. pS473-AKT was used as a marker of catalytically activated AKT1–3 because this antibody recognizes all three AKTs when phosphorylated at the cognate sites (S474-AKT2 and S472-AKT3). Consistent with the published literature, treatment of MTG001 and MTG004 cells with inhibitors of PI3'-kinase (GDC0941 or BYL719) or AKT (MK-2206 or siAKT1–3) led to decreased pS473-AKT at 22 or 24 h after treatment (Figure 3A). Treatment of cells with GDC-0068 or INY-03–041 increased phosphorylated pS473-AKT, consistent with the proposed mechanism of hyperphosphorylation, leading to a locked, inactive conformation. In cells treated with siAKT1–3, we observed substantially decreased phosphorylation levels of proteins downstream of mTORC1, including pT389-p70^{S6K}, pS235-RPS6, and pS2448-mTORC, in cell lines derived from the MTG001 or MTG004 MPDX models. Interestingly, compared to the AKT inhibitors MK2206 and GDC0068, siAKT1–3 at 24 h significantly reduced pT389-p70^{S6K} (Figure 3B, $p < 0.0001$ in both MTG001 and MTG004), pS235-RPS6 (Figure 3C), and pS2448-mTORC (Figure 3D). These results were verified by immunoblotting and compared to the respective total levels of each protein (Supplementary Figure S6B). In addition to observing consistent differences in pS473-AKT and pS235-RPS6, we also saw decreased levels of phosphorylated PRAS40 and total AKT in siAKT1–3 cells at 24 h compared to the pharmacological inhibitors in each of the MTG001 and MTG004 cell lines. Moreover, siAKT1–3 displayed highly diminished levels of 4EBP1 (pThr37/46) downstream of mTORC1. These data suggested that AKT promotes cell viability in an mTORC1-dependent manner.

Combined inhibition of AKT and SGK decreases melanoma cell proliferation

To further elucidate the role of mTORC1 and mTORC2 in this context, we evaluated whether the combined pharmacological inhibition of AKT and SGK would decrease melanoma cell proliferation. Similar to AKT, SGK1 activation depends on upstream activators PI3K and mTOR. To that end, we employed GSK650394, a pharmacological inhibitor of SGK that selectively inhibits SGK1 and 2 but has very limited inhibitory effects on AKT (29). MTG001 and MTG004 melanoma cells were treated with vehicle control, MK2206 (2.5 μ M, AKTⁱ1), GDC-0068 (1 μ M, AKTⁱ2), INY-03–041 (500nM, AKT PROTAC), GSK650394 (3 μ M, SGK inhibitor), or the combination of GSK650394 plus each of the various AKT inhibitors. As expected, treatment with AKT inhibitors did not result in a significant decrease in cell proliferation. In contrast, treatment with the SGK inhibitor GSK650394 alone resulted in a statistically significant decrease in cell proliferation. However, the cells continued to grow slowly throughout the treatment. Strikingly, treatment with the combination of the SGK inhibitor together with either the AKT inhibitors or AKT

PROTAC led to cell stasis and the absence of continued cellular proliferation (Figure 4A). Thus, although AKT inhibitors alone are ineffective, their activity is enhanced when used in combination with SGK inhibitors.

To investigate the downstream mechanisms involved in decreased melanoma cell proliferation with combined AKT and SGK inhibition, we again utilized a fluorescent bead-based quantitative immunoassay technology to measure the phosphorylation of a panel of eight cell signaling molecules in the PI3K>AKT>mTORC signaling pathway, as described above. pS473-AKT confirmed the mechanism of action of each pharmacological inhibitor. Interestingly, treatment with GSK650394 yielded a statistically significant increase in pS473-AKT (Figure 4B), just as inhibition of AKT via siAKT1–3 yielded an increase in SGK1 expression, suggesting further compensatory feedback between these two signaling proteins. Importantly, downstream phosphorylation of pT389-p70^{S6K} (Figure 4C) and pS235/236-RPS6 (Figure 4D) was significantly decreased in each of the MTG001 and MTG004 lines following combined treatment with MK2206 and GSK650394, as well as GDC-0068 and GSK650394. MK2206 and GSK650394 yielded a statistically significant decrease in p-mTOR at Ser2448 compared to either of the single-agent AKT inhibitors, MK2206 and GDC-0068, alone in MTG004 (Figure 4E). These lysates were further evaluated by immunoblotting and compared to total protein levels, showing evidence of lower levels of p-PRAS40 (Y248), p-S6 (Ser235/236), and p-4EBP1 (Thr37/46) with combination treatment versus single agent treatment, particularly in MTG001 (Figure 4F). Based on these data, we concluded that single-agent pharmacological inhibition of AKT does not effectively suppress downstream mTORC signaling, which is necessary to inhibit melanoma cell proliferation. However, the combined suppression of AKT and SGK or genetic silencing of AKT led to decreased melanoma cell growth.

Because the combination of AKT and SGK inhibition *in vitro* was quite promising, we evaluated this combination *in vivo*. Due to the toxicity reported with the use of AKT inhibitors alone (30), we first completed a maximum tolerated dose study with the combination. This combination was tolerable with minimal (less than 10% body weight) weight loss over a ten-day period and no adverse physiological or behavioral signs were noted in treated mice. We then evaluated the efficacy of this treatment regimen by analyzing its effect on *in vivo* tumor growth. Immune-competent syngeneic “glowing head” mice were injected with the YUMM3.2 mouse melanoma cell line (BRAF^{V600E}/INK4A-ARF^Δ), as well as a PTEN-deficient isogenic variant (BRAF^{V600E}/INK4A-ARF^Δ/PTEN^Δ) of this line generated using CRISPR/CAS9 technology (31). Mice bearing subcutaneous tumors were dosed five days a week with vehicle alone, GDC-0068 at 40 mg/kg, GSK650394 at 100 mg/kg, or a combination of GDC-0068 and GSK650394 (Figure 4G). Treatment with either GDC-0068 or GSK650394 did not extend overall survival compared with the vehicle. In contrast, treatment with the combination of AKT and SGK inhibition prolonged overall survival (median survival of 22 days compared with 12 days for the vehicle). This was also significantly different from AKT inhibition alone (p=0.007) and SGK inhibition alone (p=0.002).

mTOR inhibitors reduce proliferation *in vitro* and increase overall survival of mice harboring BRAF-mutant melanoma

As our analysis of siAKT1–3 knockdown and combined pharmacological inhibition of AKT and SGK suggested mTORC1 and mTORC2 dependency, we evaluated selective inhibitors of mTOR that target the catalytic-dependent functions of mTOR and are capable of inhibiting both mTORC1 and 2 as a more direct and tractable therapeutic option. Previous studies in our lab have demonstrated the inability of rapamycin, a first-generation selective mTORC1 inhibitor, to inhibit melanoma cell proliferation (32). Thus, we evaluated later-generation mTOR inhibitors, including RapaLink-1, a drug that links rapamycin and an mTOR kinase inhibitor, to inhibit both mTORC1 and 2, and paxalisib, a dual PI3K-mTOR inhibitor (21,33). Treatment with each of these pharmacological inhibitors led to a significant decrease in melanoma cell proliferation in A375, MTG001, and MTG004 cells, similar to siAKT1–3 knockdown ($p < 0.0001$ compared to DMSO control) (Figure 5A). This further suggested that cell death due to AKT knockdown was due to diminished mTOR activity.

Immunoblotting was performed to evaluate the levels of phosphorylated downstream effectors, particularly AKT, PRAS40, S6-kinase, and 4EBP1, in drug-treated cells (Figure 5B). Interestingly, both paxalisib and RapaLink led to a significant reduction in phosphorylated AKT at 16 h, but this was followed by reactivation of AKT at 24 h in both MTG001 and MTG004 cell lines. This pattern was also observed with PRAS40 phosphorylation at 16 and 24 h. However, both drugs reduced phosphorylated S6-kinase by ~50% and reduced phospho-4EBP1 to undetectable levels compared with DMSO-treated cells. This is consistent with our hypothesis that inhibition of both PI3K and mTORC1/2 effectively inhibits downstream signaling through the PI3K/AKT cascade, leading to its negative effect on melanoma cell growth.

Next, we evaluated the effect of paxalisib *in vivo*. Immunocompetent mice tolerized to eGFP and luciferase were subcutaneously implanted with YUMM 3.2 Pten^{-/-} cells, as described previously (31). Mice bearing tumors were dosed with vehicle or 15 mg/kg paxalisib once per day for five consecutive days, followed by a 2-day holiday. Strikingly, treatment with paxalisib led to inhibition of tumor growth and significantly extended the overall survival of these mice ($p = 0.0003$) (Figure 6A). There have been numerous reports of toxicities associated with dual PI3K-mTOR inhibitors (34). However, the body weights of all individual mice treated with 15 mg/kg paxalisib remained consistent without considerable loss or adverse effects throughout treatment, demonstrating that paxalisib was well tolerated.

As many patients with BRAF-mutant melanomas exhibit resistance to MAPK targeted therapies, we next sought to evaluate the effect of paxalisib on the MTG004 patient derived xenograft model that is resistant to dabrafenib and trametinib, inhibitors of mutant BRAF and MEK, respectively. Subcutaneous tumors were induced in immunodeficient mice. Once tumors were reached 100–200 mm³, the mice were treated with either vehicle, the combination of dabrafenib (30 mg/kg daily) and trametinib (1 mg/kg; five days on, two days off), or paxalisib (10 mg/kg daily) for 28 days. All mice treated with paxalisib survived without adverse toxicity for the duration of the study. Treatment with dabrafenib

and trametinib led to faster tumor growth compared with vehicle control treatment, but treatment with paxalisib significantly abrogated tumor growth (Figure 6B). Immunoblotting was performed to evaluate the levels of phosphorylated downstream effectors in drug-treated cells (Figure 6C). Paxalisib led to a significant reduction in phosphorylated AKT, PRAS40, S6-kinase, and 4EBP1. Increased levels of the apoptotic marker cleaved caspase 3 and corresponding decreased levels of total caspase 3 were also observed.

DISCUSSION

We have previously demonstrated that activated AKT can substitute for Pten loss to promote melanomagenesis *in vivo* (19). In addition, the Karreth lab has recently shown that AKT is sufficient to overcome PTEN inhibition to promote melanoma cell proliferation, low density colony formation, anchorage-independent growth, migration/invasion, and xenograft tumor growth. They further observed that PTEN suppresses melanoma primarily through inhibition of AKT (35). Despite compelling preclinical data supporting the role of AKT signaling in melanoma initiation and progression, AKT inhibitors lack efficacy in this disease (36,37). Furthermore, although there are five FDA-approved PI3'-kinase inhibitors for cancer or PI3K α -related overgrowth spectrum disorders (alpe-, copan-, duve-, idela-, and umbralisib), there are currently no FDA-approved AKT inhibitors for cancer therapy, although capivasertib in combination with faslodex has shown promising activity in breast cancer clinical trials (38,39). Hence, we sought to further elucidate the role of AKT in melanoma cell proliferation and survival. Interestingly, pharmacological inhibition of AKT had little to no effect on melanoma proliferation, but genetic suppression of AKT exhibited potent anti-proliferative activity. Moreover, the requirement for silencing of all three AKT paralogs suggests a substantial degree of overlap in AKT function for melanoma cell proliferation. Importantly, rescue of this phenotype was AKT kinase dependent.

Although AKT activation requires phosphorylation of both T308 and Ser473, the necessity of phosphorylation at each of these two residues for full activation has been debated (27). Pioneering research by Alessi et al. demonstrated that phosphorylation by both PDK1 and mTORC2 is necessary for full activation (40,41). Although maximal activation has been demonstrated to be dependent on T308 and S473 phosphorylation, AKT has been shown to be active in the absence of S473 phosphorylation (42). Based on the readout of siAKT1–3 rescue by activated AKT, phosphorylation of S473 appears largely dispensable. This is consistent with the literature describing T308 as the major regulator of Akt kinase activity; Akt phosphorylation on T308 by recombinant active PDK1 increased AKT protein kinase activity by 30-fold, whereas phosphorylation of AKT on Ser473 by DNA-dependent protein kinase (DNA-PK) only increased kinase activity by 10-fold (43). Furthermore, T308 has been described as the primary phosphorylation site by which localization of AKT to the plasma membrane allows mTORC2 to access S473 on AKT.

The majority of melanomas display phosphorylation of direct mTORC targets such as 4E-BP1 and p70^{S6K} (44) and evidence from this study shows that *BRAF*-mutated melanomas, regardless of PTEN status, rely on mTORC1 signaling. However, mTORC activation is often under the dual control of both the *BRAF*^{V600E}>MEK>ERK and PI3'-kinase signaling pathways (45). Consistent with this hypothesis, we report here that single-agent

pharmacological inhibition of AKT or SGK is insufficient to fully suppress mTORC signaling. In other cancer lines, studies have also suggested that the activity of either AKT or SGK1 is sufficient to mediate signaling through mTOR (46), which has also been observed in melanoma. AKT and SGK both phosphorylate p27, thereby blocking nuclear import and maintaining active cyclin-E/Cdk2 complexes (47,48). However, SGK is also phosphorylated directly by mTORC1. Our results suggest that in the absence of functional AKT protein, mTORC1 activity is eliminated, preventing the phosphorylation of SGK and subsequently p27.

Although our results suggest that the proliferation of melanoma cells is dependent on mTOR, phase II clinical trials of mTOR inhibitors have not shown clinical efficacy in melanoma. This may be due to multiple reasons. First, mTOR inhibitors, such as rapamycin, function by destabilizing the mTORC1-Raptor complex while leaving the mTORC2-Rictor complex intact. Rictor enables mTORC2 to directly phosphorylate S473-AKT and facilitates T308 phosphorylation by PDK1, which our data indicates is necessary for melanoma cell viability (47). As both AKT and SGK are phosphorylated by mTORC2 and PDK1 to facilitate downstream signaling through mTORC1, the residual activity of mTOR may be sufficient to drive melanoma cell proliferation. With the current inadequacy of AKT inhibitors to effectively suppress mTOR signaling, our findings suggest that the use of newer generation mTOR inhibitors, such as RapaLink, or dual mTOR/PI3K inhibitors, such as paxalisib, may be more effective. RapaLink targets both mTORC complexes, and paxalisib targets PI3K and mTOR thereby eliminating reactivation of the pathway following relief of negative feedback signaling.

Constitutive activation of PI3K/AKT activity has been well recognized in melanoma progression and has also been found in the setting of MAPK inhibitor drug resistant disease (49); thus, effective pharmacological inhibition of this pathway is highly warranted (30). Our data suggest that mTOR activity is critical for melanoma cell survival. The PI3K/mTOR inhibitor BEZ235 was previously described and tested in clinical trials; however, it failed to progress clinically owing to toxicity and lack of efficacy (50). Recently, the FDA granted orphan drug designation to paxalisib for the treatment of patients with atypical rhabdoid or teratoid tumors (AT/RT), which is a rare and aggressive childhood brain cancer. Paxalisib is also currently in clinical trials for glioblastoma and brain metastases. Our results support the use of paxalisib as a single agent either in the first line or MAPK inhibitor resistant setting for BRAF-mutant cutaneous melanoma. In this paper, we demonstrate the beneficial use of next generation PI3K/mTOR inhibitors, notably paxalisib, to inhibit melanoma cell growth. Further work by the Karreth lab shows that PTEN regulates FRA1 in an AKT-dependent manner through mTOR-mediated translation control (35). Our collective studies identify the PTEN/AKT/mTOR/FRA1 axis as an important driver of melanoma development and maintenance that could reveal additional avenues for therapeutic intervention.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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REFERENCES

1. Eroglu Z, Ribas A. Combination therapy with BRAF and MEK inhibitors for melanoma: latest evidence and place in therapy. *Ther Adv Med Oncol* 2016;8(1):48–56 doi 10.1177/1758834015616934. [PubMed: 26753005]
2. Damsky W, Micevic G, Meeth K, Muthusamy V, Curley DP, Santhanakrishnan M, et al. mTORC1 activation blocks BrfV600E-induced growth arrest but is insufficient for melanoma formation. *Cancer Cell* 2015;27(1):41–56 doi 10.1016/j.ccell.2014.11.014. [PubMed: 25584893]
3. Dankort D, Curley DP, Cartledge RA, Nelson B, Karnezis AN, Damsky WE Jr., et al. Brf(V600E) cooperates with Pten loss to induce metastatic melanoma. *Nat Genet* 2009;41(5):544–52 doi 10.1038/ng.356. [PubMed: 19282848]
4. Davies MA. The role of the PI3K-AKT pathway in melanoma. *Cancer journal* 2012;18(2):142–7 doi 10.1097/PPO.0b013e31824d448c. [PubMed: 22453015]
5. Deuker MM, McMahon M. Rational targeting of BRAF and PI3-Kinase signaling for melanoma therapy. *Mol Cell Oncol* 2016;3(3):e1033095 doi 10.1080/23723556.2015.1033095. [PubMed: 27314067]
6. Manca A, Lissia A, Capone M, Ascierto PA, Botti G, Caraco C, et al. Activating PIK3CA mutations coexist with BRAF or NRAS mutations in a limited fraction of melanomas. *J Transl Med* 2015;13:37 doi 10.1186/s12967-015-0401-8. [PubMed: 25627962]
7. Sanchez-Vega F, Mina M, Armenia J, Chatila WK, Luna A, La KC, et al. Oncogenic Signaling Pathways in The Cancer Genome Atlas. *Cell* 2018;173(2):321–37 e10 doi 10.1016/j.cell.2018.03.035. [PubMed: 29625050]
8. Kircher DA, Trombetti KA, Silvis MR, Parkman GL, Fischer GM, Angel SN, et al. AKT1(E17K) Activates Focal Adhesion Kinase and Promotes Melanoma Brain Metastasis. *Mol Cancer Res* 2019;17(9):1787–800 doi 10.1158/1541-7786.MCR-18-1372. [PubMed: 31138602]
9. Martelli AM, Tabellini G, Bressanin D, Ognibene A, Goto K, Cocco L, et al. The emerging multiple roles of nuclear Akt. *Biochim Biophys Acta* 2012;1823(12):2168–78 doi 10.1016/j.bbamcr.2012.08.017. [PubMed: 22960641]
10. Dai DL, Martinka M, Li G. Prognostic significance of activated Akt expression in melanoma: a clinicopathologic study of 292 cases. *J Clin Oncol* 2005;23(7):1473–82 doi 10.1200/JCO.2005.07.168. [PubMed: 15735123]
11. Durban VM, Deuker MM, Bosenberg MW, Phillips W, McMahon M. Differential AKT dependency displayed by mouse models of BRAF(V600E)-initiated melanoma. *J Clin Invest* 2013;123(12):5104–18 doi 10.1172/Jci69619. [PubMed: 24200692]
12. Dinavahi SS, Noory MA, Gowda R, Drabick JJ, Berg A, Neves RI, et al. Moving Synergistically Acting Drug Combinations to the Clinic by Comparing Sequential versus Simultaneous Drug Administrations. *Mol Pharmacol* 2018;93(3):190–6 doi 10.1124/mol.117.110759. [PubMed: 29242354]
13. Rozengurt E, Soares HP, Sinnet-Smith J. Suppression of Feedback Loops Mediated by PI3K/mTOR Induces Multiple Overactivation of Compensatory Pathways: An Unintended

Consequence Leading to Drug Resistance. *Mol Cancer Ther* 2014;13(11):2477–88 doi 10.1158/1535-7163.MCT-14-0330. [PubMed: 25323681]

14. Silva JM, Bulman C, McMahon M. BRAFV600E cooperates with PI3K signaling, independent of AKT, to regulate melanoma cell proliferation. *Mol Cancer Res* 2014;12(3):447–63 doi 10.1158/1541-7786.MCR-13-0224-T. [PubMed: 24425783]
15. Di Cristofano A SGK1: The Dark Side of PI3K Signaling. *Curr Top Dev Biol* 2017;123:49–71 doi 10.1016/bs.ctdb.2016.11.006. [PubMed: 28236975]
16. Di Giacomo AM, Ascierto PA, Pilla L, Santinami M, Ferrucci PF, Giannarelli D, et al. Ipilimumab and fotemustine in patients with advanced melanoma (NIBIT-M1): an open-label, single-arm phase 2 trial. *The Lancet Oncology* 2012;13(9):879–86 doi 10.1016/S1470-2045(12)70324-8. [PubMed: 22894884]
17. Eichler AF, Loeffler JS. Multidisciplinary Management of Brain Metastases. *The Oncologist* 2007;12(7):884–98 doi 10.1634/theoncologist.12-7-884. [PubMed: 17673619]
18. Loftus SK, Larson DM, Watkins-Chow D, Church DM, Pavan WJ. Generation of RCAS vectors useful for functional genomic analyses. *DNA Res* 2001;8(5):221–6 doi 10.1093/dnares/8.5.221. [PubMed: 11759842]
19. Cho JH, Robinson JP, Arave RA, Burnett WJ, Kircher DA, Chen G, et al. AKT1 Activation Promotes Development of Melanoma Metastases. *Cell reports* 2015;13(5):898–905 doi 10.1016/j.celrep.2015.09.057. [PubMed: 26565903]
20. Bromberg-White JL, Webb CP, Patacsil VS, Miranti CK, Williams BO, Holmen SL. Delivery of short hairpin RNA sequences by using a replication-competent avian retroviral vector. *J Virol* 2004;78(9):4914–6. [PubMed: 15078973]
21. Heffron TP, Ndubaku CO, Salphati L, Alicke B, Cheong J, Drobnick J, et al. Discovery of Clinical Development Candidate GDC-0084, a Brain Penetrant Inhibitor of PI3K and mTOR. *ACS Med Chem Lett* 2016;7(4):351–6 doi 10.1021/acsmchemlett.6b00005.
22. Hirai H, Sootome H, Nakatsuru Y, Miyama K, Taguchi S, Tsujioka K, et al. MK-2206, an allosteric Akt inhibitor, enhances antitumor efficacy by standard chemotherapeutic agents or molecular targeted drugs in vitro and in vivo. *Mol Cancer Ther* 2010;9(7):1956–67 doi 10.1158/1535-7163.MCT-09-1012. [PubMed: 20571069]
23. Sun L, Huang Y, Liu Y, Zhao Y, He X, Zhang L, et al. Ipatasertib, a novel Akt inhibitor, induces transcription factor FoxO3a and NF-kappaB directly regulates PUMA-dependent apoptosis. *Cell Death Dis* 2018;9(9):911 doi 10.1038/s41419-018-0943-9. [PubMed: 30185800]
24. VanBrocklin MW, Verhaegen M, Soengas MS, Holmen SL. Mitogen-activated protein kinase inhibition induces translocation of Bmf to promote apoptosis in melanoma. *Cancer Res* 2009;69(5):1985–94 doi 10.1158/0008-5472.CAN-08-3934. [PubMed: 19244105]
25. You I, Erickson EC, Donovan KA, Eleuteri NA, Fischer ES, Gray NS, et al. Discovery of an AKT Degrader with Prolonged Inhibition of Downstream Signaling. *Cell Chem Biol* 2020;27(1):66–73 e7 doi 10.1016/j.chembiol.2019.11.014. [PubMed: 31859249]
26. Vivanco I, Chen ZC, Tanos B, Oldrini B, Hsieh WY, Yannuzzi N, et al. A kinase-independent function of AKT promotes cancer cell survival. *Elife* 2014;3 doi 10.7554/eLife.03751.
27. Hart JR, Vogt PK. Phosphorylation of AKT: a mutational analysis. *Oncotarget* 2011;2(6):467–76 doi 10.18632/oncotarget.293. [PubMed: 21670491]
28. Manning BD, Toker A. AKT/PKB Signaling: Navigating the Network. *Cell* 2017;169(3):381–405 doi 10.1016/j.cell.2017.04.001. [PubMed: 28431241]
29. Sherk AB, Frigo DE, Schnackenberg CG, Bray JD, Laping NJ, Trizna W, et al. Development of a small-molecule serum- and glucocorticoid-regulated kinase-1 antagonist and its evaluation as a prostate cancer therapeutic. *Cancer Res* 2008;68(18):7475–83 doi 10.1158/0008-5472.CAN-08-1047. [PubMed: 18794135]
30. Parkman GL, Foth M, Kircher DA, Holmen SL, McMahon M. The role of PI3'-lipid signalling in melanoma initiation, progression and maintenance. *Exp Dermatol* 2022;31(1):43–56 doi 10.1111/exd.14489. [PubMed: 34717019]
31. Day CP, Carter J, Weaver Ohler Z, Bonomi C, El Meskini R, Martin P, et al. “Glowing head” mice: a genetic tool enabling reliable preclinical image-based evaluation of cancers in

- immunocompetent allografts. PLoS One 2014;9(11):e109956 doi 10.1371/journal.pone.0109956. [PubMed: 25369133]
32. Ballou LM, Lin RZ. Rapamycin and mTOR kinase inhibitors. J Chem Biol 2008;1(1–4):27–36 doi 10.1007/s12154-008-0003-5. [PubMed: 19568796]
 33. Kuroshima K, Yoshino H, Okamura S, Tsuruda M, Osako Y, Sakaguchi T, et al. Potential new therapy of Rapalink-1, a new generation mammalian target of rapamycin inhibitor, against sunitinib-resistant renal cell carcinoma. Cancer Sci 2020;111(5):1607–18 doi 10.1111/cas.14395. [PubMed: 32232883]
 34. Hung MC, Wang WP, Chi YH. AKT phosphorylation as a predictive biomarker for PI3K/mTOR dual inhibition-induced proteolytic cleavage of mTOR companion proteins in small cell lung cancer. Cell Biosci 2022;12(1):122 doi 10.1186/s13578-022-00862-y. [PubMed: 35918763]
 35. Xu X, Bok I, Jasani N, Wang K, Chadourne M, Deng O, et al. The lipid phosphatase activity of PTEN dampens FRA1 expression via AKT/mTOR signaling to suppress melanoma. bioRxiv 2023:2023.06.01.543131 doi 10.1101/2023.06.01.543131.
 36. Tolcher AW, Patnaik A, Papadopoulos KP, Rasco DW, Becerra CR, Allred AJ, et al. Phase I study of the MEK inhibitor trametinib in combination with the AKT inhibitor afuresertib in patients with solid tumors and multiple myeloma. Cancer Chemother Pharmacol 2015;75(1):183–9 doi 10.1007/s00280-014-2615-5. [PubMed: 25417902]
 37. Shoushtari AN, Ong LT, Schoder H, Singh-Kandah S, Abbate KT, Postow MA, et al. A phase 2 trial of everolimus and pasireotide long-acting release in patients with metastatic uveal melanoma. Melanoma Res 2016;26(3):272–7 doi 10.1097/CMR.0000000000000234. [PubMed: 26795274]
 38. Vanhaesebroeck B, Perry MWD, Brown JR, Andre F, Okkenhaug K. PI3K inhibitors are finally coming of age. Nat Rev Drug Discov 2021;20(10):741–69 doi 10.1038/s41573-021-00209-1. [PubMed: 34127844]
 39. Capivasertib plus Faslodex significantly improved progression-free survival vs. Faslodex in CAPitello-291 Phase III trial in advanced HR-positive breast cancer. 2022.
 40. Alessi DR, James SR, Downes CP, Holmes AB, Gaffney PR, Reese CB, et al. Characterization of a 3-phosphoinositide-dependent protein kinase which phosphorylates and activates protein kinase Balph α . Curr Biol 1997;7(4):261–9 doi 10.1016/s0960-9822(06)00122-9. [PubMed: 9094314]
 41. Alessi DR, Andjelkovic M, Caudwell B, Cron P, Morrice N, Cohen P, et al. Mechanism of activation of protein kinase B by insulin and IGF-1. EMBO J 1996;15(23):6541–51. [PubMed: 8978681]
 42. Alessi DR, Caudwell FB, Andjelkovic M, Hemmings BA, Cohen P. Molecular basis for the substrate specificity of protein kinase B; comparison with MAPKAP kinase-1 and p70 S6 kinase. FEBS Lett 1996;399(3):333–8 doi 10.1016/s0014-5793(96)01370-1. [PubMed: 8985174]
 43. Vincent EE, Elder DJ, Thomas EC, Phillips L, Morgan C, Pawade J, et al. Akt phosphorylation on Thr308 but not on Ser473 correlates with Akt protein kinase activity in human non-small cell lung cancer. Br J Cancer 2011;104(11):1755–61 doi 10.1038/bjc.2011.132. [PubMed: 21505451]
 44. Karbowniczek M, Spittle CS, Morrison T, Wu H, Henske EP. mTOR is activated in the majority of malignant melanomas. J Invest Dermatol 2008;128(4):980–7 doi 10.1038/sj.jid.5701074. [PubMed: 17914450]
 45. Mendoza MC, Er EE, Blenis J. The Ras-ERK and PI3K-mTOR pathways: cross-talk and compensation. Trends Biochem Sci 2011;36(6):320–8 doi 10.1016/j.tibs.2011.03.006. [PubMed: 21531565]
 46. Castel P, Scaltriti M. The emerging role of serum/glucocorticoid-regulated kinases in cancer. Cell Cycle 2017;16(1):5–6 doi 10.1080/15384101.2016.1232071. [PubMed: 27636373]
 47. Hong F, Larrea MD, Doughty C, Kwiatkowski DJ, Squillace R, Slingerland JM. mTOR-raptor binds and activates SGK1 to regulate p27 phosphorylation. Mol Cell 2008;30(6):701–11 doi 10.1016/j.molcel.2008.04.027. [PubMed: 18570873]
 48. Toker A mTOR and Akt signaling in cancer: SGK cycles in. Mol Cell 2008;31(1):6–8 doi 10.1016/j.molcel.2008.06.007. [PubMed: 18614042]
 49. Corrales E, Levit-Zerdoun E, Metzger P, Mertes R, Lehmann A, Munch J, et al. PI3K/AKT signaling allows for MAPK/ERK pathway independency mediating dedifferentiation-

driven treatment resistance in melanoma. *Cell Commun Signal* 2022;20(1):187 doi 10.1186/s12964-022-00989-y. [PubMed: 36434616]

50. Pongas G, Fojo T. BEZ235: When Promising Science Meets Clinical Reality. *Oncologist* 2016;21(9):1033–4 doi 10.1634/theoncologist.2016-0243. [PubMed: 27566248]

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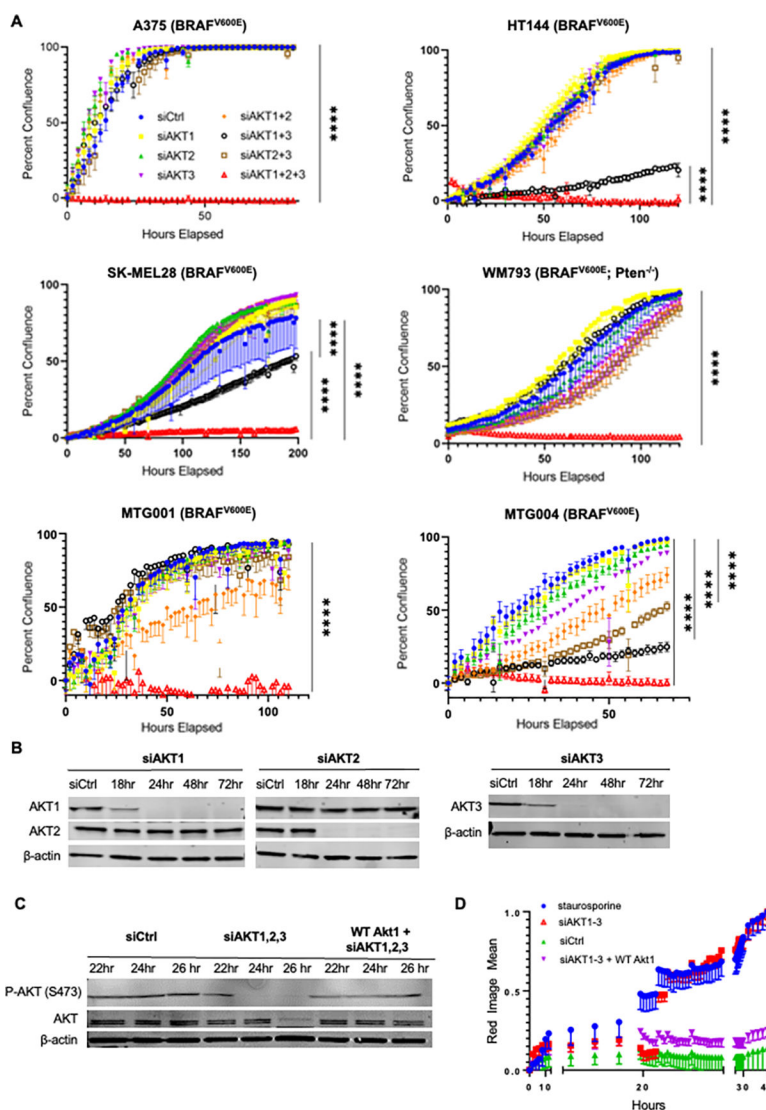


Figure 1: SiRNA-mediated knockdown of AKT1–3 inhibits proliferation and leads to cell lethality.

A, Cell confluency assay under genetic inhibition. A375, HT144, SK-MEL28, WM793, MTG001, and MTG004 cells showed little sensitivity to individual siRNAs against AKT1, 2, and 3 and AKT1+2 or AKT2+3, whereas siAKT1–3 led to complete inhibition of proliferation in all cell lines ($P < 0.0001$ in all cell lines). HT144, SK-MEL28, and MTG004 cells also exhibited sensitivity to siAKT1+3 ($P < 0.001$). Error bars indicate standard error of the mean of triplicate wells. **B**, Immunoblotting analysis of A375 cells treated with siAKT1 and siAKT2 in combination showed specific knockdown of individual paralogs using antibodies against total AKT1 and AKT2 as well as individual knockdown of AKT3 using siAKT3. **C**, Immunoblotting of A375 cells treated with siCtrl, siAKT1–3, and siAKT1–3 + overexpression of wild-type mouse Akt1 resulted in the complete knockdown of phospho-AKT (Ser473) at 24 and 26 h in siAKT1–3 vs. siCtrl, which was rescued by the addition of mouse Akt1. Total AKT protein levels were also significantly reduced at 26 h by siAKT1–3. **D**, Cell death assay comparing cells treated with staurosporine, siCtrl, siAKT1–

3, and siAKT1–3 + WT Akt1. Treatment with siAKT1–3 increased cell death around 22–24 h post-transfection, comparable to the level of staurosporine, a multi-kinase inhibitor, as a control for cell death, which was completely rescued by the addition of mouse WT Akt1.

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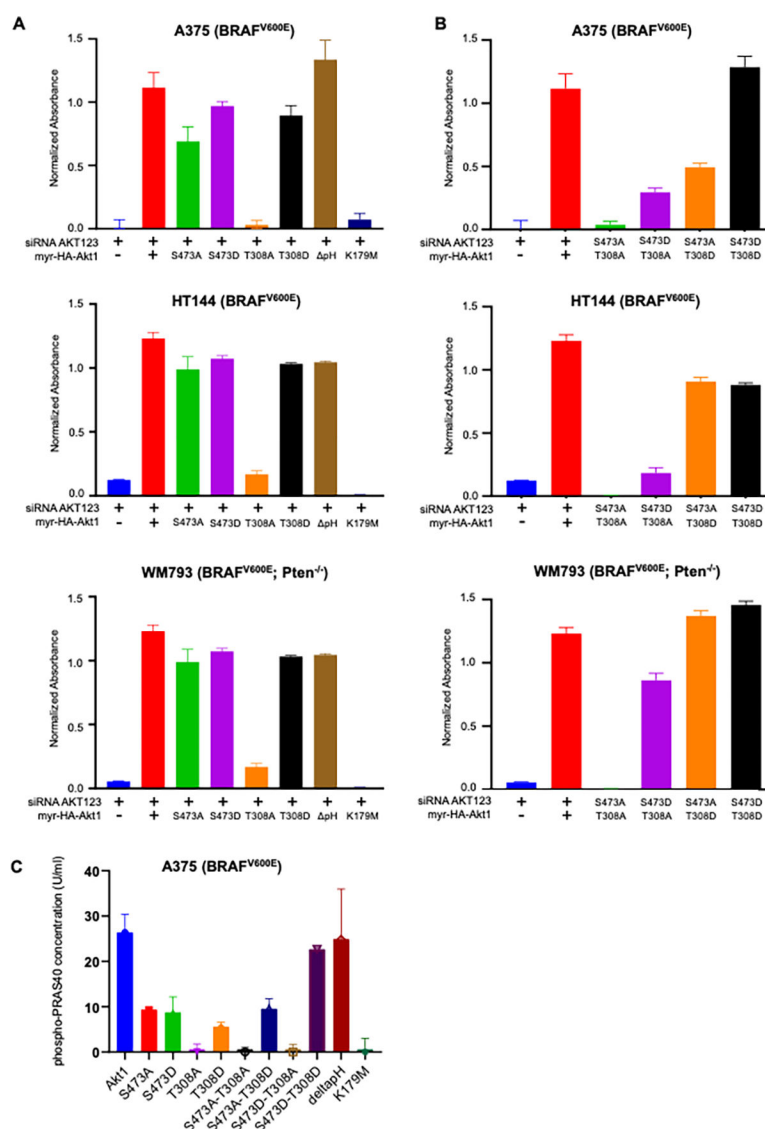


Figure 2: Rescue of melanoma cells from siRNA-mediated knockdown of AKT1–3 is dependent on Akt kinase activity and T308 phosphorylation.

A, MTT assay of stable A375, HT144, and WM793 cell lines expressing phospho- and kinase mutants after 48 h of siAKT1–3 transfection. Cell lines expressing T308A or K179M mutations were unable to significantly rescue siAKT1–3 knockdown, whereas all other cells expressing phospho mutants were able to rescue this phenotype. **B**, MTT assay of stable A375, HT144, and WM793 cell lines expressing double phospho- and kinase mutants after 48 h of siAKT1–3 transfection. **C**, P-PRAS40 kinase ELISA of A375 myrAkt1 stable cell lines expressing phospho- and kinase mutants T308A, S473A, T308A, S473D, T308A, and K179M demonstrated undetectable levels of P-PRAS40.

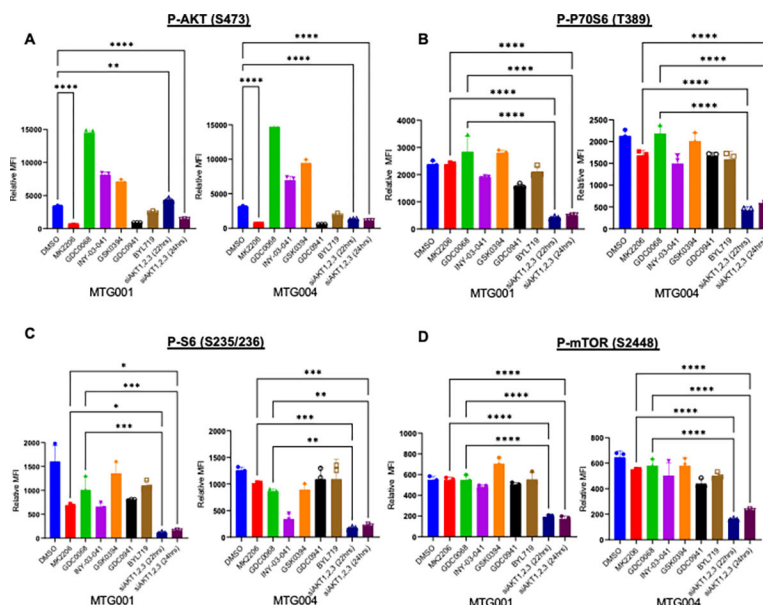


Figure 3: SiAKT1–3 significantly decreases mTORC activity.

Luminex quantitative immunoassay of phosphorylation of AKT (Ser473), p70S6 (Thr389), S6 (Ser235/236), and mTOR (Ser2448). MTG001, and MTG004 were treated with DMSO control, MK2206 (2.5 μ M), GDC0068 (1 μ M), INY-03–041 (500nM), GDC0941 (1 μ M), BYL719 (1 μ M), siAKT1–3 at 22 hours (50nM), or siAKT1–3 at 24 hours (50nM). **A**, MK2206, GDC0941, and siAKT1–3 (24 h) significantly decreased P-AKT ($P < 0.0001$ in both cell lines) compared to the DMSO control. SiAKT1–3 (22 h) significantly decreased P-AKT compared to the DMSO control in MTG004 ($P < 0.0001$). GDC0068, INY-03–041, and GSK0394 significantly increased P-AKT levels compared with the DMSO control ($P < 0.0001$ for all treatments). **B**, SiAKT1–3 at 22 and 24 h significantly decreased P-p70S6 compared to MK2206 and GDC0068 in both MTG001 and MTG004 ($P < 0.0001$). **C**, SiAKT1–3 (22 h) and siAKT1–3 (24 h) decreased P-S6 in comparison to MK2206 ($P = 0.02$ and $P = 0.04$, respectively) and GDC0068 in MTG001 ($P = 0.0002$ and 0.0004 , respectively). SiAKT1–3 (22 h) and siAKT1–3 (24 h) decreased P-S6 in comparison to MK2206 ($P = 0.0001$ and 0.0002 , respectively) and GDC0068 ($P = 0.0011$ and 0.0020 , respectively) in MTG004. **D**, SiAKT1–3 at 22 and 24 h significantly decreased P-mTOR compared to MK2206 and GDC0068 in both MTG001 and MTG004 ($P < 0.0001$).

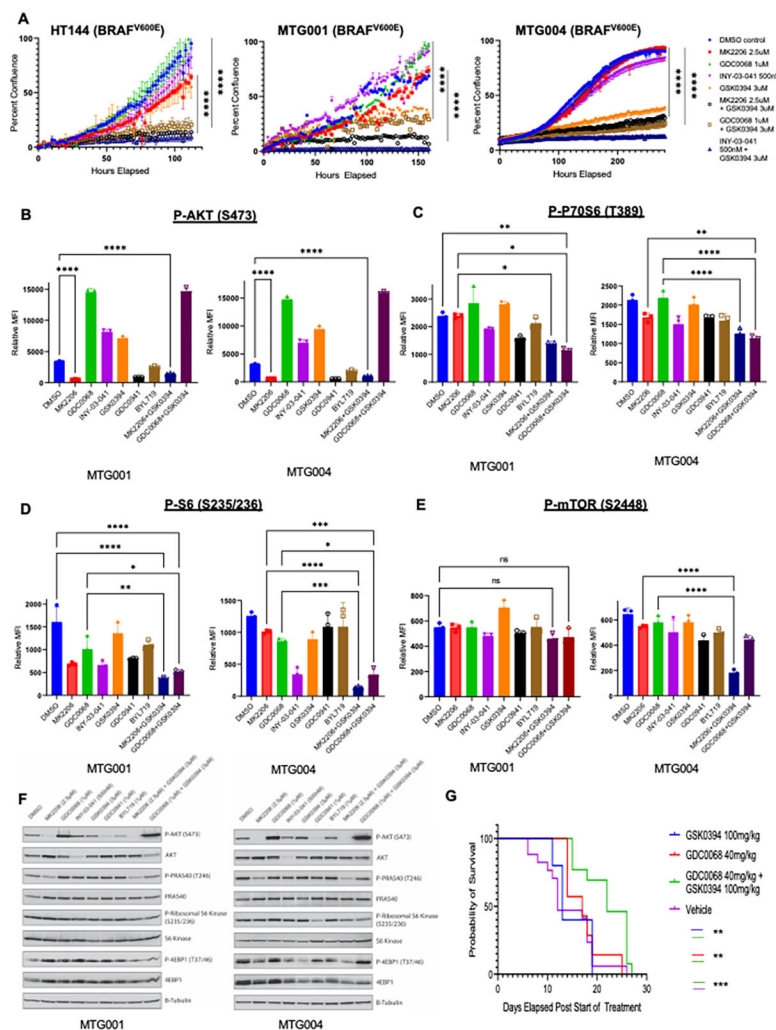


Figure 4: Combined inhibition of AKT and SGK decreases melanoma cell proliferation through effects on mTORC signaling.

A, Cell confluency assay under pharmacological inhibition. HT144, MT001, and MTG004 cells were treated with DMSO control, MK2206 (2.5μM), GDC0068 (1μM), INY-03-041 (500nM), GSK0394 (3μM), MK2206 (2.5μM) + GSK0394 (3μM), GDC0068 (1μM) + GSK0394 (3μM), and INY-03-041 (500nM) + GSK0394 (3μM). Treatment with the combination of MK2206 + GSK0394, GDC0068 + GSK0394, and INY-03-041 + GSK0394 led to a significant decrease in cell proliferation in all three cell lines ($P < 0.0001$). Treatment with GSK0394 alone also led to a statistically significant decrease in cell proliferation in MTG001 and MTG004 cells ($P < 0.0001$). **B-E**, Luminex quantitative immunoassay of phosphorylation of AKT (Ser473), p70S6 (Thr389), P-S6 (Ser235/236A), and mTOR (Ser2448). **B**, MK2206, GDC0941, and MK2206 + GSK0394 significantly decreased P-AKT ($P < 0.0001$), whereas GDC0068, GSK0394, INY-03-041, and GDC0068 + GSK0394 significantly increased P-AKT ($P < 0.0001$) in both MTG001 and MTG004 cells. **C**, The double combination of MK2206 + GSK0394 or GDC0068 + GSK0394 significantly decreased P-p70S6 levels compared to MK2206 ($P = 0.042$ and $P = 0.0124$, respectively) in MTG001. GDC0068 + GSK0394 significantly decreased P-p70S6 compared

with either a single AKT inhibitor (MK2206 or GDC0068) ($P < 0.0001$) in MTG004. **D**, Both combination treatments (MK2206 + GSK0394 and GDC0068 + GSK0394) decreased P-S6 compared to GDC0068 alone in MTG001 ($P = 0.0039$ and $P = 0.0468$, respectively). Combination treatments decreased P-S6 compared to both single AKT agents alone in MTG004 cells (vs. MK2206: $P < 0.0001$ and $P = 0.0008$, vs. GDC0068: $P = 0.0004$ and 0.0110 , respectively). **E**, There was no significant decrease in mTOR phosphorylation by combination treatment versus single-agent inhibition in MTG001, but MK2206 + GSK0394 resulted in a significant decrease in P-mTOR versus either single AKT agent alone in MTG004 ($P < 0.0001$). **F**, Immunoblotting analysis of lysates treated with pharmacological inhibitors for 24 hours and used for Luminex analysis (B-E). Decreased levels of P-PRAS40, P-S6, and P-4EBP1 in combination treatments versus single agents alone were observed, particularly in MTG001. **G**, Kaplan-meier survival curve of immunocompetent mice tolerized to luciferase and GFP were injected subcutaneously with YUMM 3.2 BRAF^{V600E}; *Cdkn2a*^{-/-}; *Pten*^{-/-} mouse melanoma cells and treated q.d. with vehicle, 100 mg/kg GSK0394, 40 mg/kg GDC0068, or the combination of GSK0394 plus GDC0068. Treatment with either GDC-0068 or GSK650394 did not extend overall survival compared to the vehicle, while treatment with combination of GDC-0068 and GSK650394 significantly prolonged overall survival.

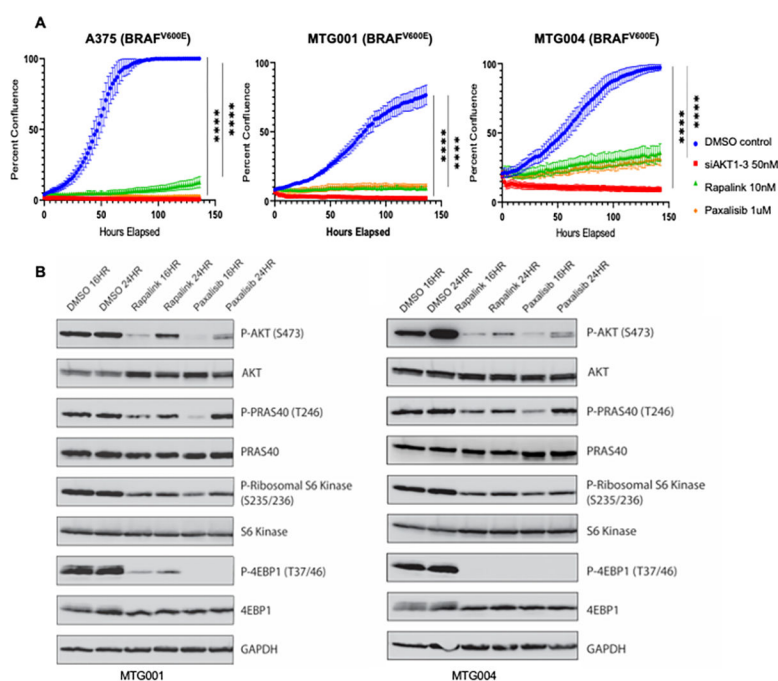


Figure 5: mTOR inhibitors reduce proliferation *in vitro*.

A, Cell confluence assays under genetic and pharmacological inhibition. A375, MTG001, and MTG004 cells were treated with the DMSO control, siAKT1–3 (50nM), RapaLink (10nM), and paxalisib (1μM). Treatment with siAKT1–3, RapaLink, and paxalisib significantly decreased cell proliferation (as measured by confluence) ($P < 0.0001$). **B**, Immunoblotting analysis of tumor lysates from mice treated with vehicle or the indicated pharmacological agent and used for MTT analysis in A.

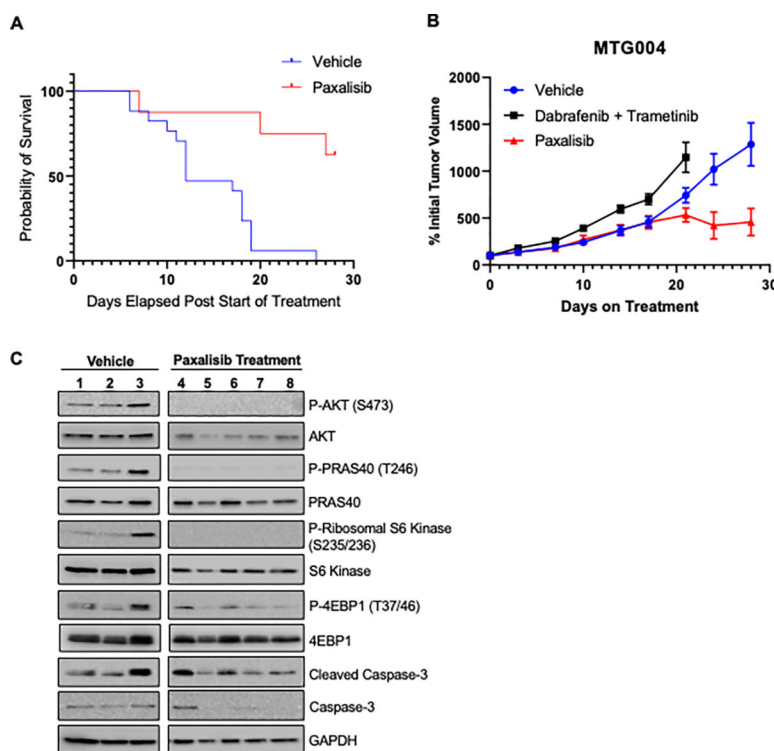


Figure 6: mTOR inhibitors increase overall survival of mice harboring BRAF-mutant melanoma.

A, Kaplan-Meier survival curve: Immunocompetent mice tolerized to luciferase and GFP were injected subcutaneously with YUMM 3.2 BRAF^{V600E};Cdkn2a^{-/-};Pten^{-/-} mouse melanoma cells treated with vehicle or 15 mg/kg paxalisib. Statistical analysis was performed using the log rank Mantel-Cox test. **B**, Tumor volumes of vehicle or treated cohorts of dabrafenib/trametinib resistant patient-derived xenografts. MTG004 tumor chunks were implanted subcutaneously into NRG mice. Once tumors reached 100–200 mm³, mice were randomized to receive vehicle (n=8), dabrafenib 30mg/kg + trametinib 1 mg/kg (n=6), or paxalisib 10 mg/kg (n=6) for 28 days. Mean tumor volume is shown for each cohort. Error bars represent standard error of the mean. P-values were determined by two-tailed student's t-test. At day 21, the mean tumor volume in the paxalisib treated mice was significantly lower than vehicle (P = .05) or dabrafenib/trametinib treated mice (P = .01). Likewise, at day 28 the mean tumor volume in the paxalisib treated mice was significantly lower than vehicle (P = .02). (C) Immunoblotting analysis of tumor lysates from mice treated with vehicle (1–3) or paxalisib (4–8).